

LESSON 5-4 • SESSION 1 • GUIDED NOTES**Area of Regular Polygons**

Learning target – I can find the area of a regular polygon by finding the area of one triangle and multiplying. · 6.G.A.1

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
regular polygon <i>polígono regular</i>	All sides _____ and all angles equal.	stop sign, hexagon tile
congruent <i>congruente</i>	Exactly the same _____ and shape.	all center triangles match
apothem <i>apotema</i>	Distance from the _____ to the middle of a side.	height of a center triangle

KEY IDEA*complete each sentence*

- Lines from the center to each vertex make _____ triangles.
- A regular polygon with n sides makes _____ center triangles.
- Area = $n \times (\frac{1}{2} \times \text{side} \times \text{_____})$.

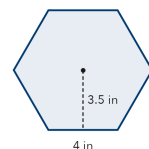
WORKED EXAMPLE*fill in the blanks as we solve*

A regular pentagon has sides of 10 cm and an apothem of about 7 cm. Find its area.

Step 1 5 sides make _____ center triangles.**Step 2** One triangle: $\frac{1}{2} \times 10 \times 7 = \text{_____ cm}^2$.**Step 3** All triangles: $5 \times 35 = \text{_____ cm}^2$.**Answer** Area $\approx \text{_____ cm}^2$ **YOU TRY***use the example as a model*

- Find the area of the regular hexagon.

Show your work.

**Answer** _____ in²

- How many center triangles does each regular polygon make?

a. octagon _____**b.** decagon _____**SUM IT UP***in your own words*

To find the area of a regular polygon, I _____ and then _____.

LESSON 5-4 • SESSION 2 • GUIDED NOTES

Composite Figures

Learning target – I can find the area of a composite figure by adding or subtracting the areas of simpler shapes. · 6.G.A.1

WORDS TO KNOW

fill in the missing words

WORD	WHAT IT MEANS	EXAMPLE
composite figure <i>figura compuesta</i>	A shape made of two or more _____ shapes.	rectangle + triangle
decompose <i>descomponer</i>	To _____ a shape into smaller shapes you know.	L shape: 2 rectangles
subtract <i>restar</i>	To _____ the area of a piece that is cut out.	wall – window

KEY IDEA

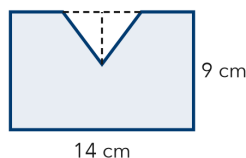
complete each sentence

- Split a composite figure into rectangles, triangles, or other shapes you _____.
- _____ the areas of pieces that are joined together.
- _____ the area of a piece that is cut out.

WORKED EXAMPLE

fill in the blanks as we solve

A rectangle is 14 cm by 9 cm. A triangle notch with base 6 cm and height 4 cm is cut out. Find the area left.



Step 1 Whole rectangle: $14 \times 9 = \underline{\hspace{1cm}} \text{ cm}^2$.

Step 2 Cut-out triangle: $\frac{1}{2} \times 6 \times 4 = \underline{\hspace{1cm}} \text{ cm}^2$.

Step 3 The triangle is removed, so _____.

Answer $126 - 12 = \underline{\hspace{1cm}} \text{ cm}^2$

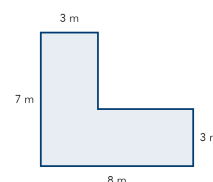
YOU TRY

use the example as a model

1. Find the area of the L-shaped figure. All corners are right angles.

Show your work.

Answer _____ m^2



2. Write add or subtract.

a. a yard, not counting its pool: _____

b. a wall plus its triangle roof: _____

SUM IT UP

in your own words

I add areas when _____, and I subtract areas when _____.
